

Do Sirtuins Promote Mammalian Longevity?: A Critical Review on Its Relevance to the Longevity Effect Induced by Calorie Restriction

Sirtuins (SIRT), a family of nicotinamide adenine dinucleotide (NAD)-dependent deacetylases, are emerging as key molecules that regulate aging and age-related diseases including cancers, metabolic disorders, and neurodegenerative diseases. Seven isoforms of SIRT (SIRT1–7) have been identified in mammals. SIRT1 and 6, mainly localized in the nucleus, regulate transcription of genes and DNA repair. SIRT3 in the mitochondria regulates mitochondrial bioenergetics. Initial studies in yeasts, nematodes, and flies indicated a strong connection of SIRT with the life-prolonging effects of calorie restriction (CR), a robust experimental intervention for longevity in a range of organisms. However, subsequent studies reported controversial findings regarding SIRT roles in the effect of CR. This review describes the functional roles of mammalian SIRTs and discusses their relevance to mechanisms underlying the longevity effect of CR.